

Instructions

1. Implement the following Mapper Algorithm. Select a clustering algorithm of your choice (K-means, DBSCAN, etc.).
2. You are provided with the file “noisy-annuli-bridge.csv”. The data is two annuli connected by a bridge. Visualize the data.
3. Apply your **Implemented Algorithm** on the above dataset, and check that you recover the topology of the original dataset.

Algorithm 1 Mapper Algorithm

Require: Data set $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$
Require: Lens (filter) function $f : X \rightarrow \mathbb{R}$
Require: Cover $\mathcal{U} = \{U_1, \dots, U_m\}$ of $f(X)$ with overlap
Require: Clustering algorithm $\mathcal{C}$
Ensure: Simplicial graph $G = (V, E)$

- 1: Initialize empty vertex set $V \leftarrow \emptyset$
- 2: Initialize empty edge set $E \leftarrow \emptyset$
- 3: **for** each cover element $U_i \in \mathcal{U}$ **do**
- 4: Compute pullback $X_i \leftarrow f^{-1}(U_i)$
- 5: Cluster X_i using $\mathcal{C}$ to obtain clusters
- 6: **for** each cluster C_{ij} **do**
- 7: Create vertex v_{ij} corresponding to C_{ij}
- 8: Add v_{ij} to V
- 9: **end for**
- 10: **end for**
- 11: **for** each pair of vertices $(v_{ij}, v_{pq}) \in V \times V$ **do**
- 12: **if** the corresponding clusters intersect nontrivially **then**
- 13: Add edge (v_{ij}, v_{pq}) to E
- 14: **end if**
- 15: **end for**
- 16: **return** $G = (V, E)$
